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Forecasting of water discharge using atmospheric and hydrologic sensors to identify long-term high-risk periods in Herkimer County, NY

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Introduction

Herkimer County is located in Upstate New York that includes part of the Mohawk River that flows across the south part of the county. From the dates of July 28th, 2013 to July 2nd 2013, many areas of Herkimer County experienced severe flash flooding due mostly to heavy rainfall. This flood caused damage to homes, businesses, roadways, bridges, and caused one fatality throughout different towns in Herkimer County (Eisenstadt, 2013). Due to its proximity to the Mohawk River and other sources of water such as streams, this area is highly prone to flooding, and public authorities as well as residents and businesses are in need of an advanced forecasting system. Previous research has been conducted in this area using spatial analysis to identify vulnerable areas of flooding (Swan et al, 2013). In this research, we utilize available atmospheric and hydrogeological time series data from available sensors near Little Falls, Herkimer County to design a forecasting algorithm for a long-term prediction of flooding in Herkimer County. Little Falls, NY was chosen as it is considered as one of the most dynamic study locations because it provides continuous daily and hourly monitoring of atmospheric and hydrological sensors along the Mohawk River to explore forecasting of flood high-risk periods.

Methodology

The USGS National Water Information System website was used to locate a surface water site with the most sufficient data. We chose the gaging station USGS 1347000 in Little Falls, NY and our data for water discharge rate and daily statistics was obtained between February 1st, 2012 and February 1st 2018. A map was created on Google Earth to highlight the stations that we utilized to retrieve data. Additionally, atmospheric data was obtained through NOAA in order to correlate with the water discharge. The raw data was retrieved through KNIME, a data mining software. Data pre-processing such filtering, replacing missing data, converting hourly data to daily data, and finding maximum and minimum values that took place through data mining simple techniques. Then, the Kolmogorov-Zurbenko filter with a length of the filter equal with 31 was applied to all variables following previous described methodology (Tsakiri et al., 2014) to derive the long-term component. Then, the processed data, which we call “long-term component”, was imported into SPSS for further advanced statistical analysis, such as forecasting. In SPSS, bivariate analysis allowed us to select correlated variables to water discharge, and a linear regression model was obtained by isolating the water discharge as the dependent variable. The independent variables used in the model are minimum temperature, maximum temperature, maximum Relative Humidity, Snow Depth, Precipitation, Maximum Atmospheric Pressure, Maximum Groundwater Depth, and Snowfall. The following section describes the results of the linear regression model.



Figure 1: A map created on Google Earth of the Greater Herkimer County area displaying the Mohawk River (flowing west to east) depositing into the Hudson River, the horizontal scale in meters, and the yellow pins on the map that represent the study area and the sensors locations from where weather stations and water wells data were obtained from (Albany International Airport Weather Station GHCND:USW00014735, Weather Data Station WBAN 04741, USGS gage station id 01346000, USGS Well 4301210745230001).

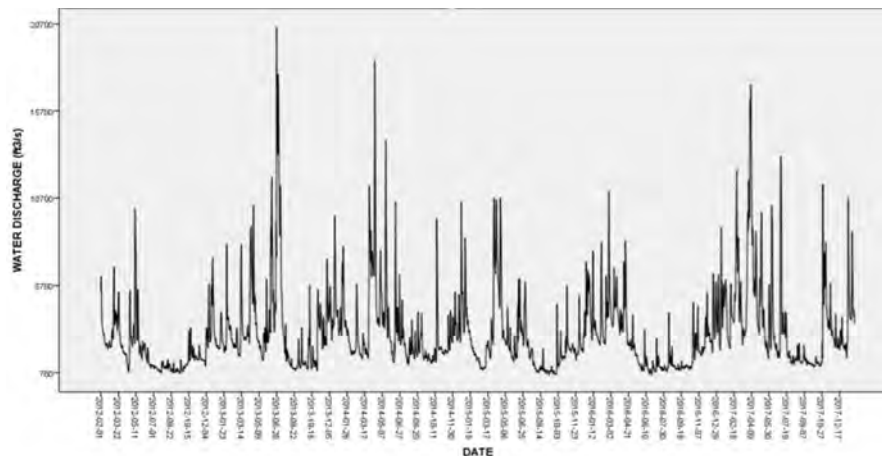


Figure 2: Time Series Plot of the raw daily water discharge data from January 2012 to February 2018. The high peaks of the time series plot indicate minor and major flooding events.

Results

Before applying the linear regression model, we examine the correlations between the variables. For the raw data, the correlation between the water discharge and the depth (groundwater) is 0.734, while the correlation between the water discharge and maximum temperature is 0.173. After the application of the Kolmogorov-Zurbenko filter, the correlation between water discharge and depth is 0.923, and the correlation between water discharge and maximum temperature is .504. Using the linear regression model, the forecasting equation has been designed after consideration of the acceptable p-values for the coefficients of the linear regression model (p-values are less than 0.05). Because all the coefficients of the linear regression model have a p value less than 0.05, we can conclude that the coefficients are statistically significant. The value of the R^2 , the coefficient of determination, of the linear regression model is equal to 0.941. Thus, 94.1% of the variation of the long term component in water discharge can be explained by the atmospheric variables and the hydrological variables. To verify the assumptions of the linear regression model, we plot the residuals of the model.

A scatterplot of the residuals, a normal probability P-P plot, and a line chart showing the long-term water discharge versus the predicted values are plotted in Figure 3. The plots show that the residuals have normal distribution and they show no significant pattern or extreme values. In addition, the variance of the residuals is constant since they are randomly distributed around zero. The predicted and the long term water discharge values seem to overlap which is reasonable due to high correlation. In the plot, the most recent peak represents heavy rainfall, which caused a flood in July 2017.

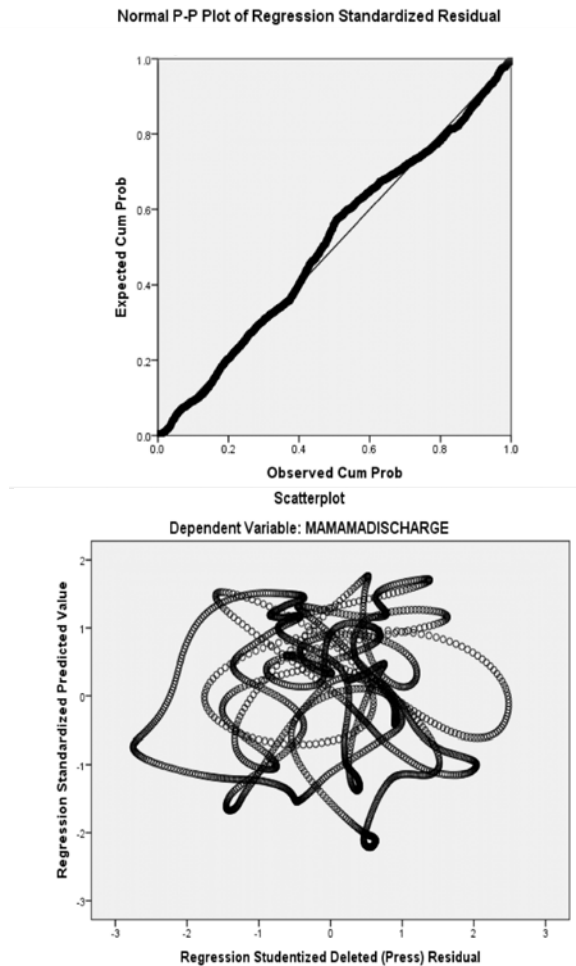


Figure 3: On the left side, a Normal Probability Plot shows the distribution of the residuals derived by the linear regression of the Water Discharge and the independent variables. On the right side, the scatter plot shows the standardized residuals versus the standardized predicted values.

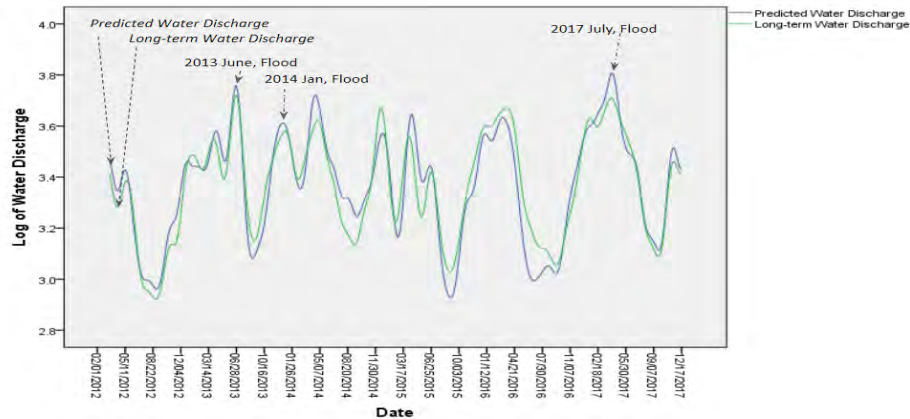


Figure 4: Time Series Plot of the predicted Values of the water discharge (green) compared with the long-term water discharge (blue). The peaks represent major flood events, such as the destructive flood that occurred on June 28, 2013 caused by persistent heavy rainfall.

Discussion

The normal probability plot (Fig. 3) shows that the data is approximately normal because most of the points fit on or around the line. All the coefficients of the linear regression model are statistically significant since the P-values are less than 0.05, and the R^2 value is 0.941 for the long term components. The R-square value on the raw data using the linear regression model is 0.886. Thus for the long term components, 94.1% of the variation in water discharge can be explained by the independent variables (from linear regression) providing a very accurate long-term forecasting as the predicted values of the water discharge almost line up with the measured values by the USGS (Fig. 4). For the long term component of the water discharge the most correlated variable is the long term component of the groundwater level. A scatterplot of the residuals (Fig. 3) displays that the data shows no linearity or outliers, indicating that there is no influence of extreme points on the results and the linear regression model is appropriate for the analysis.

Long term data were retrieved through statistical processing in KNIME and the Kolmogorov-Zurbenko filtering to eliminate short term variations (considered as noise). The correlation between the water discharge and the maximum temperature significantly increased for the long term components. The correlation between the water discharge and groundwater increased by approximately 25.7%. The correlation between the water discharge and maximum temperature increased by approximately three times. The purpose of this analysis is to have the ability to predict long term trends. Although these trends will not have the ability to predict a flood within a few days, it can predict monthly flooding.

Conclusion

The results indicate that the long-term data can be used to predict flood patterns over a long period of time, while the raw data do not produce as significant results. Therefore, in order to predict floods in spans of months, the long-term data should be used to justify the flood prediction. This forecasting model will allow to identify high-risk flood periods of time in advance, which is important for citizens to be prepared for such geohazard from ahead of time to ensure their safety.

References:

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Poster Presentation